

ARTIFICIAL INTELLIGENCE	
Course Code: ISE552	Credits: 3:0:0
Pre – requisites: Nil	Contact Hours:42
Course Coordinator: Dr Jagadeesh Sai D	

Course Contents

Unit 1

Introduction to Artificial Intelligence, definition and goals of AI, foundations of AI, types of AI, applications of AI in engineering and society.

Intelligent Agents: Agents and environments, sensors and actuators, rational agents, rationality, performance measures, PEAS representation. single-agent and multi-agent environments.

Types of Agents: Simple reflex agents, model-based reflex agents, goal-based agents, utility-based agents, learning agents.

Pedagogy: Conceptual Learning through NPTEL video lectures and discussion-based sessions

Links: 1. <https://nptel.ac.in/courses/112103280> 2. <https://nptel.ac.in/courses/106106126>

3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod01lec02.mp4>

Unit 2

Problem-Solving Agents: Problem formulation, initial state, actions, transition model, goal test, path cost, state-space representation. Uninformed Search: Breadth First Search, Depth First Search, Uniform Cost Search, Depth-Limited Search, Iterative Deepening Search. Informed Search: Heuristic search, heuristic functions, Greedy Best-First Search, A* Search, admissible heuristics, consistent heuristics. Local Search: Hill Climbing, Simulated Annealing, local search applications. Adversarial Search: Game playing, game formulation, Minimax algorithm, Alpha-Beta pruning, evaluation functions.

Pedagogy: Case-based Learning using NPTEL videos and real-world case studies

Links: 1. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod03lec08.mp4>

2. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod03lec09.mp4>

3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod04lec11.mp4>

Unit 3

Knowledge-Based Agents: Knowledge and intelligence, knowledge representation, knowledge bases, inference. Propositional Logic: Propositions, logical connectives, truth tables, logical equivalence, inference, resolution. First-Order Logic: Predicates, variables, constants, functions, quantifiers, representation of knowledge using First-Order Logic. Reasoning: Forward chaining, backward chaining, rule-based reasoning.

Pedagogy: Hands-on Learning with NPTEL tutorials and guided agents building

Links: 1. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod08lec23.mp4>

2. <https://nptel.ac.in/courses/106106140>

3. <https://media.dev.nptel.ac.in/content/mp4/112/103/112103280/MP4/mod04lec10.mp4>

Unit 4

Uncertainty in AI: Sources of uncertainty, probability basics, conditional probability, Bayes' theorem, Bayesian reasoning, introduction to Bayesian Networks.

Generative AI: Introduction to Generative AI, generative versus discriminative AI, Large Language Models, Transformer architecture – high-level understanding, foundation models, multimodal AI, AI assistants.

Recommended Resources / Links: <https://www.coursera.org/learn/raspberry-pi-platform>

Unit 5

Prompt Engineering: Introduction to prompting, zero-shot prompting, few-shot prompting, role prompting, structured prompting, prompt evaluation, hallucinations and limitations of Generative AI.

Responsible and Ethical AI: AI bias, fairness, transparency, Explainability, privacy, security, copyright and intellectual property, human oversight, responsible use of Generative AI.

AI Applications: AI in healthcare, cybersecurity, finance, smart cities, robotics, education, manufacturing and software engineering.

Recommended Resources / Links:

1. <https://media.dev.nptel.ac.in/content/mp4/107/106/107106090/MP4/mod01lec03.mp4>
2. <https://nptel.ac.in/courses/106105077>
3. <https://media.dev.nptel.ac.in/content/mp4/106/102/106102220/MP4/mod12lec92.mp4>

Text book

Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson Education, 2021/2022

Reference books

1. Pradeep Singh, Tapan K. Gandhi and Balasubramanian Raman, *Artificial Intelligence and Machine Learning*, 1st Edition, McGraw Hill India, 2025
2. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd Edition, O'Reilly Media, 2022

Self-Learning: Project Based Learning

Course Outcomes:

At the end of the course, the student will be able to:

1. Understand the fundamentals of Artificial Intelligence and apply problem-solving techniques using agents to solve the problems. (PO-1,2,3,5,12; PSO-1,3)
2. Analyze uninformed and informed search strategies in solving real world problems. (PO -1,2,3,5,12; PSO-1,3)
3. Apply adversarial search methods to make optimal game decisions and construct knowledge-based agents using logical reasoning. (PO- 1, 2,3,5,12; PSO-1,3)
4. Apply knowledge using the syntax and semantics of First-Order Logic, and inference techniques to solve reasoning problems. (PO- 1, 2,3,5,6,12; PSO-1,3).
5. Apply algorithms for classical planning in AI and analyze the philosophical, ethical, and safety implications of artificial intelligence. (PO- 1, 2,3,5,8, 12; PSO-1,3)

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	30	CO1, CO2, CO3
Internal Test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other Components		
MCQ/ Project	20	CO1, CO2, CO3, CO4,CO5
Semester-End Examination	100 (Scale down to 50 Marks)	CO1, CO2, CO3, CO4, CO5